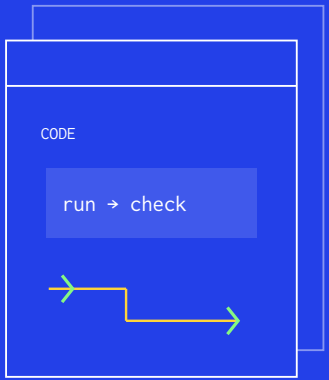
WEEK 05 / PROJECT CLINIC & PRESENTATIONS

MLOps: Machine Learning Operations

Project clinic & team presentations

Final preparation, end-to-end system presentations, final submission and the course grade.



Session plan

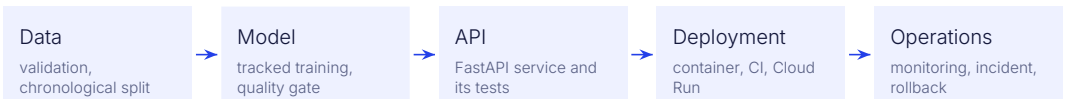
Four hours: preparation first, then presentations, with a break and a buffer between them.

Test your laptop on the projector during the break. The slot buffer absorbs setup and small overruns.

Preparation time: three tasks

The presentation follows your system end to end, as it actually runs.

YOUR PRESENTATION FOLLOWS THE SYSTEM



Show it running: live demo on your laptop; code where it explains a decision.

1 **Complete and verify**

Finish open work. Rerun the checks on the revision you will present.

2 **Prepare the 10 minutes**

Slides and a live demo that follow the chain above, from data to recovery.

3 **Rehearse once**

Time it. Every member must be able to explain any part of the system.

Final submission: one end-to-end system

The Release wraps up what you learned and built in Weeks 1–4, as one system that runs.

W1 Git, data, baseline | W2 container, API | W3 deployment, logs, CI | W4 gates, incidents → Release v1.0.0

1

Finish on release/v1.0.0

Apply final fixes, including feedback from today.

2

Merge to main, tag v1.0.0

The tag marks the exact revision that is graded.

3

Publish Release v1.0.0

From the tag, with notes and attached files (next slide).

4

Check access

Instructor minhtc-uca can open the repo and the Release.

DEADLINE	Sunday 22 November 2026, 23:59
WHERE	`github.com/<owner>/mlops-<session>-<team>-bike-demand/releases`

What Release v1.0.0 contains

Three places on one Release page. File paths are fixed; the content is your team's work.

`v1.0.0` **Release v1.0.0 · Bike Demand** — tag v1.0.0 on main

IN THE REPOSITORY AT THE TAG

`README.md`
How to run, test and deploy the service

`RUNBOOK.md`
Recovery procedure, at the repository root

`configs/`
Configuration and gate rules

`src/  tests/`
Code, unit/API/container tests, CI workflow

`reports/`
Incident record and gate decisions

RELEASE NOTES (TEXT ON THE PAGE)

Identities
Commit, model version + sha256, image tag

Final evaluation
One held-test result of the selected model

Verification
CI run links, deployment and /version output

Known limitations
What is incomplete, and why

ASSETS (ATTACHED FILES)

`model bundle + .sha256`
The accepted model: artifacts/ is not in Git

`final-presentation.pdf`
Final presentation slides

Course grade: project 50%, final exam 50%

Two equal parts: what your team built and presented, and what you understand individually.

50% FINAL SYSTEM + LAB REPORTS
Project

WITHIN THE PROJECT: 60 / 40, AS IN THE SYLLABUS

60% Final system

Release v1.0.0, presentation, live demo, Q&A.
→ 30% of the course grade

40% Lab reports

Weeks 1–4, individual.
→ 20% of the course grade

50% INDIVIDUAL
Final exam (MCQ)

- Multiple choice: one answer per question
- Date and logistics: announced by the school

Final exam: revision topics

A reminder from lectures and labs, not a list of questions. Some questions may go a little further.

EXAM FORMAT Multiple choice · choose exactly one answer per question

01 MLOps & CI/CD

MLOps lifecycle · reproducibility · CI, CD, CT · automated tests · deployment pipelines · dev, staging, prod · configuration, environment variables · secrets · Infrastructure as Code

02 Data & ML pipelines

data cleaning, preprocessing · formats, serialization · relational, document, key-value, graph · data validation · train, validation, test · leakage · scikit-learn Pipeline · DAGs, task dependencies

03 Experiments & models

training, evaluation · metrics · hyperparameters · MLflow experiments, runs, parameters, artifacts · model registry, versions · evaluation gate · promotion · reproducible runs

04 Docker & containers

images, containers · Dockerfile · layers · containers vs VMs · shared kernel · portability · port mapping · environment configuration · registries · Kubernetes basics, orchestration

05 APIs & serving

REST · HTTP methods, status codes · FastAPI, Pydantic validation · stateless APIs · microservices vs monolith · API gateway · clean architecture · online vs batch prediction · latency, throughput

06 Testing & quality

unit vs integration tests · pytest · API tests · regression tests · tolerances · flaky tests · tests in CI · Arrange–Act–Assert · coverage

07 Git, Actions & cloud

Git, Gitflow, pull requests · GitHub Actions: workflows, jobs, steps, triggers · reading CI failures · IaaS, PaaS, SaaS · Cloud Run · Artifact Registry · IAM · cloud cost · SageMaker, Vertex AI, Azure ML

08 Monitoring & scaling

structured logs · health checks · p50, p95, p99 · resource utilization · data, prediction drift · dashboards, alerts · rollback · pruning, quantisation, distillation · batching · distributed training

NEXT

Team preparation: 1 h 45 – 2 h
Presentations after the break

Test your laptop on the projector during the break.
Final submission: Release v1.0.0 by Sunday 22 November 2026, 23:59.